higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

17 August 2023 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists 6 pages and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a new page.
5. Use only a blue or black pen.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Define the following terms:

1.1.1 Offset

1.1.2 Vena contracta

1.1.3 Absorption

1.1.4 Poise

1.1.5 Density

(5 × 2) (10)

1.2 Give ONE term for each of the following descriptions by writing it next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 The temperature of fluid when condensation takes place

1.2.2 Operates on the principle that when a conductor cuts a magnetic a field, an emf is induced in the conductor

1.2.3 When these fluids are deformed by strain, the ratio between shear rate and shear stress will be a constant value

1.2.4 Tapping situated 25,4 mm from the upstream and 25,4 mm from the downstream face of the orifice plate

1.2.5 A control loop where a correction is made after an error has been detected

(5 × 1) (5)

1.3 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 An electromagnetic flow meter is capable of measuring the volumetric flow rate of non-conductivity liquids.

1.3.2 The Reynolds number is a dimensionless criterion that expresses the ratio of inertial forces to viscosity forces in a fluid flowing in a closed system.

1.3.3 An open-loop control system does not need a human operator to take corrective measures.

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1.3.4 The relationship between the pipe diameter and the diameter of the hole in the orifice plate cannot influence the calibration of a gauge.

1.3.5 When the number of hydrogen ions exceeds the number of hydroxyl ions, the solution is acidic.

(5 × 1) (5)
[20]

TOTAL SECTION A: 20

SECTION B

QUESTION 2: FLOW MEASUREMENTS

2.1 Draw neat sketches that will match the descriptions given below.

2.1.1 It is a circular metal plate with a central hole with the circumference of the plate. (2)

2.1.2 It is the irregular movement of a fluid. It is made up of many smaller currents swirling and moving in all directions. (2)

2.1.3 It is the position where the velocity of the flow is the highest and the pressure is the lowest after passing through a constriction (2)

2.2 A V-notch is installed in a channel to maintain a flow rate of 0,7 m^3/s. If the V has an angle of 50°, calculate the head developed in the weir. (5)

2.3 State THREE advantages and TWO disadvantages of a drag body flowmeter. (5)

2.4 What is the difference between a *target flow meter* and a *drag body flowmeter*? (2)

2.5 State FOUR factors of importance when selecting a turbine flow meter for any given application. (4)
[22]

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QUESTION 3: DENSITY, HUMIDITY AND VISCOSITY

3.1 Name THREE methods used in liquid density measurements. (3)

3.2 Explain, with the aid of a fully labelled sketch, the operation of a capacitive hygrometer for solids. (7)

3.3

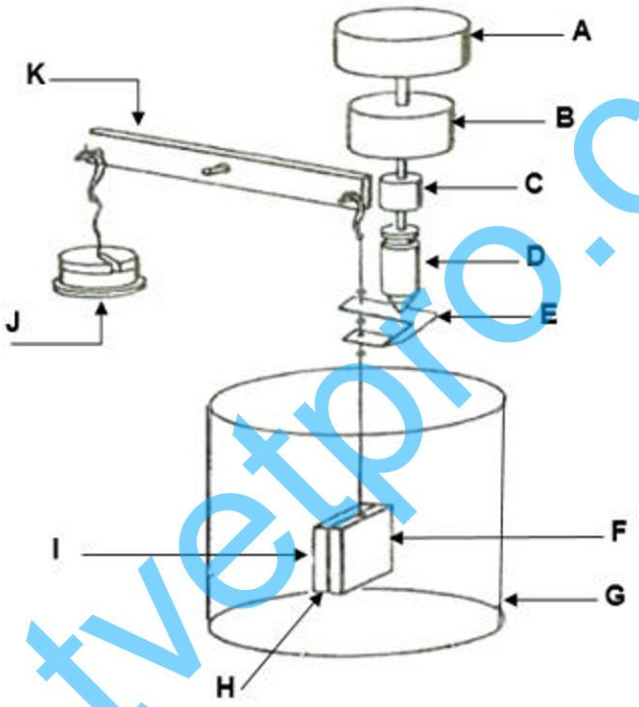


FIGURE 1

3.3.1 Name the instrument in FIGURE 1 above. (2)

3.3.2 Identify components A–K. (11)
[23]

QUESTION 4: pH AND CONDUCTIVITY MEASUREMENT

4.1 The pH of swimming pool water was found to be 6,85. Calculate the amount of hydrogen ions [H^+] present. (4)

4.2 What is the basic difference between *conductivity* and *pH*? (2)

4.3 How would you overcome change in temperature of the conductivity of a solution? (4)

4.4 Make a neat labelled sketch for pH solution potential measurement. (5)
[15]

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QUESTION 5: AUTOMATIC CONTROL

5.1 Refer to FIGURE 2 below and answer the questions that follow.

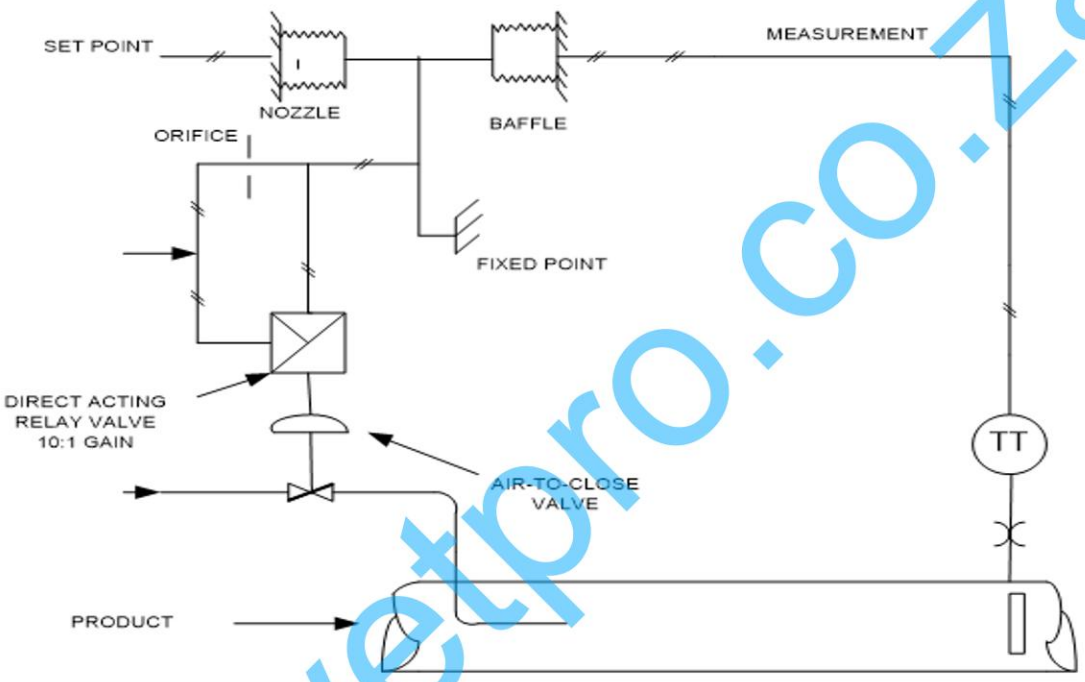


FIGURE 2

5.1.1 Name the device in the figure above. (2)

5.1.2 Describe the operation of this device. (7)

5.2 Make a neat labelled sketch of a force-balance proportional action controller. (7)

5.3 Name FOUR forms of discrimination in electronic control. (4)
[20]

TOTAL SECTION B: 80
GRAND TOTAL: 100

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INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$$Q = \frac{8}{15} \quad Tan \frac{\theta}{2} \sqrt{2g.H^5}$$

$$Q = \frac{2}{3} \; B\,C\sqrt{2g.H^3}$$

$$Q = \frac{2}{3} \; CB\sqrt{2g.H^3}$$

pH = -log[H^+]

pOH = -log [OH^-]

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